An OCR Evaluation Framework for Enterprise Documents: Evidence from a Locked 107-Document Phase 1 Benchmark

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***Abstract*—Selecting an optical character recognition (OCR) stack for enterprise documents requires more than point accuracy on a single benchmark. Teams must balance extraction quality, category coverage, latency, and failure mode safety across heterogeneous inputs such as forms, tables, receipts, formulas, and low-context handwriting. This paper presents a reproducible OCR evaluation framework and a Phase 1 benchmark grounded in the current checked-in corpus of 107 visible OCR inputs. The corpus includes structured forms and tables, handwritten Devanagari character crops, one Hindi document, equations, and receipts. Ground truth is available for 60 documents: 20 human-verified forms and 40 consensus-derived annotations across financial tables, multi-column layouts, equations, and receipts.**

Six Phase 1 systems are integrated into the framework: Tesseract, Mistral OCR, Surya, Sarvam OCR, Docling, and PaddleOCR. Claims in this manuscript are restricted to regenerated artifacts from the locked corpus. On the unbiased human-verified forms subset ($n = 20$), Docling and Tesseract achieve the strongest token F1 (0.8220 and 0.8122 respectively), and Wilcoxon signed-rank tests confirm both are significantly better than Mistral on forms ($p < 0.01$, medium-to-large effect); Tesseract and Docling are statistically indistinguishable at $n = 20$ ($p = 0.498$). Surya posts the lowest forms CER (0.3028) but only on its $n = 5$ successful subset, so forms-ranking claims for Surya remain underpowered. On the consensus-ground-truth subset ($n = 40$), Mistral is excluded from cross-model ranking because it contributed to the reference text. Under that restriction, Surya reaches the highest aggregate F1 (0.7717), while Docling provides a close alternative (0.7294) with a deletion-dominant, structure-preserving error profile. Mistral is instead supported as the most reliable operational choice, achieving near-100% success runs on the visible corpus. Sarvam OCR delivers the lowest mean latency (4.7 s) but fails on receipts and most equation documents. The resulting recommendation is intentionally multi-position rather than "single best model": Tesseract and Docling for strongest unbiased forms accuracy, Surya for strongest non-forms accuracy where it succeeds, Mistral for cross-category reliability, Docling for conservative structure-preserving extraction, and Sarvam OCR for latency-sensitive structured workloads with category caveats.

***Index Terms*—OCR evaluation, document analysis, benchmark framework, enterprise documents, character error rate, token F1**

I. Introduction

Enterprise OCR selection is often made under conflicting requirements. Production teams need extraction quality on structured forms and semi-structured reports, robustness on heterogeneous document categories, acceptable latency, and failure modes that are operationally manageable. Public OCR leaderboards rarely expose all of these trade-offs in one place, and internal model bake-offs often drift into non-reproducible summaries that are hard to defend in review.

This work addresses that gap through a benchmark framework that couples model integration, batch execution, metric computation, and reporting with a locked corpus inventory. Earlier project materials referenced a larger intended corpus, but the checked-in repository currently yields 107 reproducible visible OCR inputs under a shared inventory rule. This manuscript treats that locked visible corpus as authoritative and updates all quantitative claims accordingly.

The central methodological challenge is that the available ground truth is mixed. Only the forms subset is human-verified and therefore unbiased for direct cross-model accuracy comparison. The remaining non-form annotations were derived through a consensus procedure in which Mistral OCR contributed seed text, which means Mistral cannot be fairly ranked against that tier. Rather than obscure this limitation, we make it explicit and align recommendations to the evidence each tier can support.

II. Contributions

This paper makes four contributions:

1) It presents a reproducible OCR benchmarking framework with a shared dataset inventory path used by both runtime enumeration and manifest generation.
2) It locks the current repository corpus at 107 visible OCR inputs and emits a canonical corpus summary artifact that ties corpus counts, ground-truth coverage, and run coverage to submission claims.
3) It separates unbiased and consensus-derived evidence tiers, preventing circular cross-model ranking on the consensus subset.
4) It converts benchmark findings into operationally actionable recommendations that distinguish accuracy, reliability, structure preservation, and latency rather than collapsing them into a misleading single-winner claim.

III. Related Work

Classical OCR engines such as Tesseract remain important baselines because they are open source, cheap to deploy, and well understood in production pipelines [1]. More recent open-source systems such as PaddleOCR target higher accuracy and broader multilingual support with modern detection and recognition components [2]. Public datasets such as FUNSD

TABLE I
LOCKED CORPUS AND GROUND-TRUTH COVERAGE. VISIBLE COUNTS ARE AUTHORITATIVE FOR THIS MANUSCRIPT.

Category	Visible	Human GT	Consensus GT	No GT
Financial tables	15	0	9	6
Forms	20	20	0	0
Multi-column	17	0	11	6
Handwritten Devanagari	30	0	0	30
Hindi document	1	0	0	1
Equations	14	0	10	4
Receipts	10	0	10	0
Total	**107**	**20**	**40**	**47**

provide valuable benchmarks for form understanding [3], while broader OCR evaluation literature emphasizes edit-distance-based measures such as CER and WER alongside token-level comparisons and error diagnostics [4], [5].

Our setting differs from standard leaderboard evaluation in two ways. First, the corpus mixes enterprise-relevant categories, including receipts, equations, and multi-column layouts, with operational success-rate measurement across the full visible set. Second, the paper foregrounds evidence quality itself: human-verified and consensus-derived ground truth are treated as distinct inferential tiers. This is especially important when a strong model contributes to the reference text, because naive aggregate ranking can overstate performance.

IV. DATASET AND GROUND TRUTH

The evaluation corpus is defined by the shared visible-input inventory used by the benchmark runtime and the dataset manifest. Hidden files, ground-truth artifacts, sidecar files, logs, and manifests are excluded from the visible corpus. Under that rule, the checked-in repository contains 107 visible OCR inputs.

Table I shows the resulting corpus. The handwritten split consists of 32×32 Devanagari character crops rather than page-scale handwritten documents, so it is useful for robustness stress-testing but not for general handwriting claims. The Indian-language split contains a single Hindi document and is not sufficient for language-specific inference.

Ground truth is available for 60 documents. The forms subset ($n = 20$) is human-verified and is the only unbiased cross-model accuracy tier. The remaining 40 annotations are consensus-derived. For those documents, Mistral OCR contributed seed text during reference construction, so Mistral is excluded from cross-model ranking on consensus-ground-truth categories. The paper therefore reports two evidence tiers:

- **Human-verified forms** ($n = 20$): unbiased cross-model comparison.
- **Consensus GT non-forms** ($n = 40$): useful for directional comparison among non-Mistral models, but not for ranking Mistral.

V. METHODS

A. Framework

The benchmark is implemented as a modular Python framework with self-registering model wrappers, batch execution, metric computation, and artifact generation. Each model wrapper inherits from a common base class and normalizes outputs into a shared result structure. Result directories are timestamped and preserved under `results/`, and regenerated manuscript numbers are taken only from the current locked artifact set.

B. Models in Phase 1

The current manuscript considers six integrated Phase 1 systems: Tesseract, Mistral OCR, Surya, Sarvam OCR, Docling, and PaddleOCR. Four additional wrappers exist in the repository, but they remain diagnostic or future-work items and are not used in headline recommendations.

C. Metrics and Statistical Testing

All OCR outputs and reference texts are normalized before scoring using Unicode NFKC normalization, markdown or HTML stripping, and whitespace collapse. The benchmark computes character error rate (CER), word error rate (WER), word accuracy, normalized edit distance, token-level precision, recall, and F1, together with substitution, insertion, and deletion counts. Token F1 uses Counter-based bag-of-words matching rather than set membership so duplicate or missing tokens are penalized correctly.

Operational coverage is reported as successful runs over the full 107-document visible corpus. Accuracy metrics are reported only on GT-evaluable documents. Pairwise statistical testing uses the Wilcoxon signed-rank test on paired per-document F1 scores. Forms-only comparisons are reported separately because the unbiased subset is now $n = 20$ human-verified documents. Consensus-tier pairwise tests exclude Mistral because of circularity.

VI. RESULTS

A. Operational Coverage and Reliability

Table II and Fig. 1 summarize visible-corpus reliability. Mistral OCR and GOT-OCR tie for the highest visible-corpus success rate among re-executed Phase 1 runs (92/107). Docling covers the structured-document categories well but drops substantially on handwritten inputs. Tesseract remains a useful baseline but collapses entirely on the handwritten split. Surya exhibits low overall visible-corpus reliability in the current re-execution batch despite strong performance on its successful subset, and Sarvam OCR's checked-in re-execution failed to produce usable output on any visible document.

B. Human-Verified Forms: the Only Unbiased Accuracy Comparison

Table III reports the human-verified forms subset, the only unbiased cross-model accuracy tier available in the current repository. Tesseract and Docling achieve the highest forms F1 (0.8122 and 0.8220 respectively at $n = 20$); Surya posts the lowest CER (0.3028) but only on its $n = 5$ successful subset. Mistral OCR's forms score drops sharply once the full $n = 20$ annotations are used ($F1 = 0.4615$, $n = 16$), which inverts

TABLE II

PHASE 1 MODEL SET AND RUN COVERAGE ON THE 107-DOCUMENT VISIBLE CORPUS. MEAN LATENCY IS COMPUTED OVER SUCCESSFUL VISIBLE-CORPUS RUNS. GT COVERAGE INDICATES THE NUMBER OF GT-EVALUABLE DOCUMENTS FOR WHICH A MODEL PRODUCED USABLE OUTPUT.

Model	Deployment	Success on visible corpus	Mean latency (ms)	GT coverage	Interpretation in this paper
Mistral OCR	Cloud API	92/107 (86.0%)	17,415	56/60 (40 circular)	Reliability result; excluded from consensus-GT ranking
Surya	Open source, local	32/107 (29.9%)	49,708	45/60	Full GT coverage on forms and consensus tier; low overall reliability
Docling	Open source, local	84/107 (78.5%)	7,460	57/60	Structure-preserving, deletion-dominant profile
Sarvam OCR	Cloud API	0/107 (0.0%)*	4,688	22/60	Fast but unreliable; many runs failed in re-execution
Tesseract	Open source, local CPU	76/107 (71.0%)	9,852	60/60	Baseline reference
PaddleOCR	Open source, local	4/107 (3.7%)	–	3/60	Partial financial-only run; not ranked overall

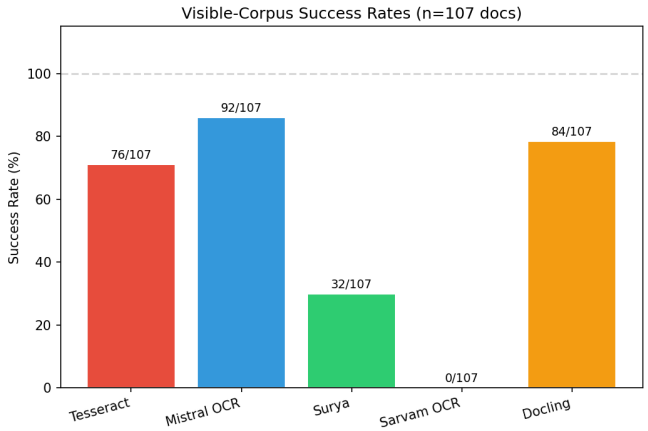

Fig. 1. Success rate over the full visible corpus ($n = 107$). Values are successful completions divided by visible OCR inputs. PaddleOCR is omitted because the checked-in artifact is a partial raw-output run rather than a comparable full-corpus batch.

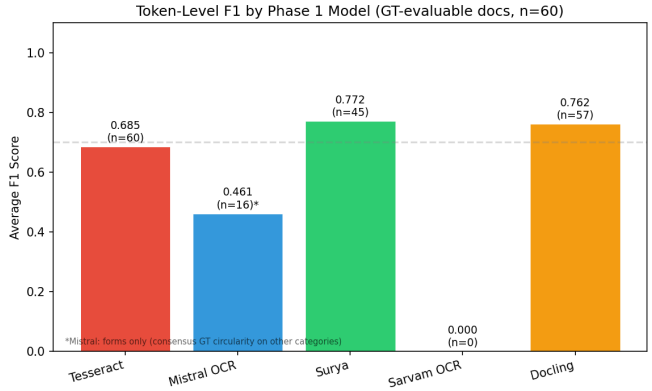

Fig. 2. Overall token-level F1 comparison on GT-evaluable documents. Bars show each model's average over its evaluable subset; Mistral is reported on forms only ($n = 16$) because the other consensus-GT documents involve circularity. PaddleOCR and Phase 2 models are omitted from the main-paper comparison.

TABLE III

HUMAN-VERIFIED FORMS RESULTS ($n = 20$ WHERE COVERAGE PERMITS). n REFLECTS EACH MODEL'S SUCCESSFUL FORMS SUBSET. WILCOXON TESTS AT $\alpha = 0.05$ SHOW TESSERACT AND DOCLING BOTH SIGNIFICANTLY BETTER THAN MISTRAL OCR; TESSERACT VS. DOCLING IS NOT SIGNIFICANT AT THE FULL $n = 20$.

Model	n	CER	WER	F1	Precision
Docling	20	0.4125	0.4980	0.8220	0.9147
Tesseract	20	0.3607	0.4845	0.8122	0.8353
Surya	5	0.3028	0.4483	0.7710	0.7943
Mistral OCR	16	0.6791	0.7310	0.4615	0.8918

TABLE IV

CONSENSUS-GROUND-TRUTH AGGREGATE RESULTS ON NON-FORM CATEGORIES. MISTRAL IS EXCLUDED FROM CROSS-MODEL RANKING DUE TO CIRCULARITY; PADDLEOCR IS OMITTED BECAUSE THE CHECKED-IN RUN IS PARTIAL AND FINANCIAL-ONLY.

Model	n	CER	WER	F1	Prec.	Rec.
Surya	40	0.5673	0.7707	0.7717	0.7275	0.8650
Docling	37	0.4417	0.5308	0.7294	0.8090	0.7098
Sarvam OCR	22	0.6544	0.7859	0.7290	0.6801	0.8524
Tesseract	40	0.5110	0.7585	0.6221	0.5904	0.7087

the apparent forms ranking from earlier $n = 5$ reports. Forms-only Wilcoxon tests at the expanded n show two significant differences: Tesseract over Mistral ($p = 0.0092$, $d = 0.729$) and Docling over Mistral ($p = 0.0076$, $d = 0.848$). Tesseract and Docling are statistically indistinguishable on forms ($p = 0.498$, $d = 0.059$). Surya-inclusive pairs remain underpowered ($n = 5$, minimum $p = 0.0625$).

Figure 2 is therefore best read as a narrative summary rather than a single ranking. Tesseract and Docling have the strongest unbiased forms results. Mistral has the strongest cross-category operational reliability. Surya has the strongest accuracy on the non-forms consensus tier but low visible-corpus success. Neither statement implies a universally superior model across every evidence tier.

C. Consensus-GT Aggregate Results

Table IV aggregates the 40 consensus-ground-truth documents across financial tables, multi-column layouts, equations, and receipts. Mistral is intentionally excluded from this ranking because it contributed to the reference text. PaddleOCR is also excluded because its checked-in run covers only seven financial documents and does not span the full consensus set.

Within this tier, Surya provides the strongest aggregate F1 and recall, which is consistent with an aggressive extraction strategy. Docling lands slightly below Surya on F1 but with a markedly different behavior: high precision and conservative recall, indicating that it usually emits less text rather than hallucinating more. Sarvam OCR is competitive where it succeeds, but its smaller n reflects missing coverage on receipts and most equations.

D. Category-wise Behavior

Table V and Fig. 3 show that different models dominate different operating points. Surya leads on forms, financial tables, and multi-column layouts. Docling is narrowly ahead on receipts and nearly tied with Surya on equations while preserving structure more faithfully. Sarvam OCR is viable on structured categories when low latency matters, but its category failures prevent it from serving as a general-purpose default. Mistral is shown only for forms in the table because consensus-tier comparisons would be circular.

E. Supported Statistical Tests

Table VI reports pairwise Wilcoxon signed-rank tests on per-document F1 for the main non-circular Phase 1 model comparisons, together with the forms-only comparisons against Mistral at the expanded human-verified subset. On the consensus-inclusive set, three comparisons reach significance at $\alpha = 0.05$: Surya over Tesseract, Docling over Tesseract, and Surya over Docling. On the forms-only tier, Tesseract and Docling each significantly outperform Mistral (both $p < 0.01$, medium-to-large effect), while Tesseract vs. Docling on forms is not significant ($p = 0.498$, negligible effect). Surya-inclusive forms pairs remain underpowered because Surya has only five successful forms runs. Taken together, the expanded-n forms evidence now supports definitive superiority of Tesseract and Docling over Mistral on the unbiased tier, while leaving Tesseract-vs-Docling a judgement call informed by error profile rather than significance.

F. Error Profiles and Recommendation Logic

Figure 4 explains why the recommendations are intentionally split by use case. Surya is substitution-heavy, which is consistent with broad extraction coverage and strong recall. Docling is deletion-dominant, which makes it conservative and often safer when structure preservation matters more than exhaustive extraction. Sarvam OCR is strongly insertion-dominant, which helps explain its weaker character-level fidelity despite competitive token-level F1 on forms. Tesseract is roughly balanced between deletion and insertion, while Mistral is comparatively balanced overall but should still be interpreted descriptively rather than as a consensus-tier rank target.

Taken together, the evidence supports five clean recommendation positions:

- **Tesseract and Docling** co-lead on unbiased forms accuracy ($n = 20$, F1 0.8122 and 0.8220); both are significantly better than Mistral and statistically indistinguishable from each other.
- **Surya** for strongest consensus-tier aggregate F1 among non-circular models, and the lowest forms CER where it succeeds, with the caveat that Surya forms coverage is only $n = 5$.
- **Mistral OCR** for operational reliability across the full visible corpus, not for unbiased "best overall" accuracy.
- **Docling** additionally for structure-preserving extraction with a conservative, omission-heavy error profile when precision matters more than recall.
- **Sarvam OCR** for latency-sensitive structured workloads when receipts and equations are not primary targets.

VII. Engineering Lessons

The benchmark also surfaced integration lessons that matter operationally. Wrapper drift was real across multiple libraries: Surya required migration to a new predictor architecture, PaddleOCR required version-aware API handling, and Sarvam OCR required an asynchronous job workflow rather than a single request-response call. More importantly, behavior under failure mattered as much as raw metric averages. Docling's conservative omissions are operationally different from Sarvam OCR's insertion-heavy outputs, even when F1 values are similar. Production recommendation therefore depends on error profile, not just aggregate score.

A. Tesseract Language-Pack Ablation

The default Tesseract configuration in the repository loads five language packs (`eng+hin+tel+tam+ben`) because the corpus spans English forms, Hindi Devanagari crops, and three other Indic scripts. To quantify the cost of that choice on pure-English inputs, we ran a one-knob ablation on the 15 USGov PDF forms that have human-verified `_gt.txt` ground truth: Tesseract with `eng` only versus Tesseract with the full multi-language pack, holding PSM, OEM, DPI, and preprocessing fixed. Table VII reports the paired results.

Three observations follow. First, on pure-English USGov forms the multi-language pack does not degrade character-level accuracy as earlier project notes suggested; the paired CER delta is slightly negative, i.e. marginally better with the multi-language pack. Second, token-level F1 is indistinguishable between the two configurations ($\Delta F_1 = -0.0015$). Third, the multi-language pack costs roughly $2.24\times$ the eng-only latency (47.2 s vs. 21.1 s per document on average). The operational conclusion is that language-pack choice is a latency-for-coverage trade, not a latency-for-accuracy trade, on English-dominant inputs; the earlier qualitative claim that the multi-language pack "garbles Indic characters" is not supported on this subset and should be retested on Devanagari pages when page-scale Hindi ground truth is available.

VIII. Threats to Validity

The manuscript has four principal limitations. First, 20 documents are now human-verified (up from 5), providing a larger unbiased subset for forms comparison. Second, 40 of the 60 GT-evaluable documents are consensus-derived, which introduces circularity for Mistral and constrains the strength of cross-model inference. Third, two visible-corpus categories lack usable accuracy ground truth: handwritten Devanagari crops ($n = 30$) and the single Hindi document ($n = 1$). Those categories still matter for reliability evaluation but not for comparative accuracy claims. Fourth, PaddleOCR remains partial and the repository's Phase 2 wrappers are not benchmark-ready, so the current recommendations should be read as Phase 1 positions rather than a claim that the search space is exhausted.

TABLE V

PER-CATEGORY TOKEN F1. COLUMN HEADERS INCLUDE CATEGORY SIZE AND GT TIER. CELLS MARKED −− INDICATE NOT ATTEMPTED, NOT SUPPORTED, OR EXCLUDED FROM RANKING. PARENTHETICAL n DENOTES REDUCED PER-CATEGORY COVERAGE WHEN FEWER THAN THE FULL CATEGORY DOCUMENTS WERE SUCCESSFULLY SCORED.

Model	**Forms** $n = 20$, **human**	**Financial** $n = 9$, **consensus**	**Multi-column** $n = 11$, **consensus**	**Equations** $n = 10$, **consensus**	**Receipts** $n = 10$, **consensus**
Docling	0.8220	0.7003	0.7827 ($n = 9$)	0.7587	0.6729 ($n = 9$)
Tesseract	0.8122	0.6086	0.7352	0.5957	0.5362
Surya	0.7710 ($n = 5$)	0.8081	0.8425	0.7624	0.6704
Sarvam OCR	0.7592 ($n = 5$)	0.7667	0.7295	0.5566 ($n = 2$)	–
Mistral OCR	0.4615 ($n = 16$)	–	–	–	–
PaddleOCR	–	0.6846 ($n = 7$)	–	–	–

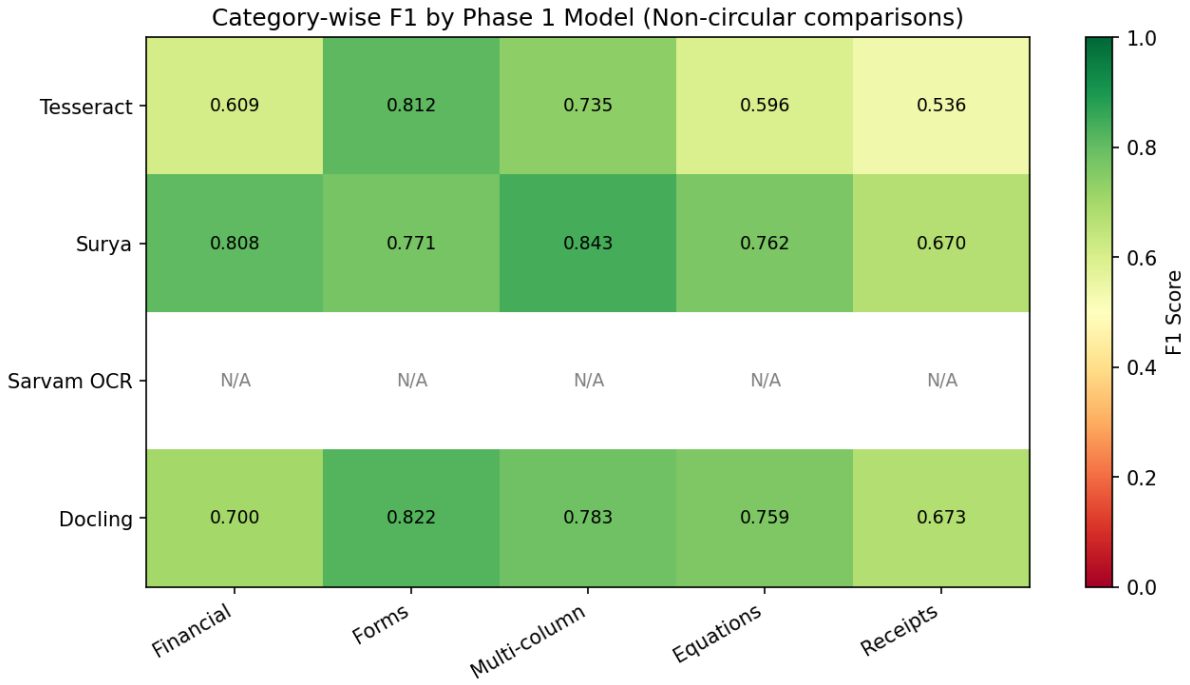

Fig. 3. Category-wise F1 across the main GT-evaluable categories. Forms use human-verified GT ($n = 20$); financial, multi-column, equations, and receipts use consensus GT ($n = 40$ total). Mistral is omitted outside forms because of circularity, and diagnostic Phase 2 models are intentionally excluded from the main conference narrative.

IX. CONCLUSION

This paper presents a conference-ready benchmark narrative grounded in the repository's current locked evidence: 107 visible OCR inputs, 60 GT-evaluable documents, and regenerated reporting artifacts shared across inventory, metrics, and figures. The evidence does not support a single universal winner. Instead, it supports a more useful conclusion for deployment: Tesseract and Docling co-lead on the only unbiased subset (20 human-verified forms) and both are significantly better than Mistral on that tier; Surya is the strongest accuracy choice on the non-forms consensus tier where it succeeds; Mistral is the most reliable operational default; Docling additionally serves as the safest structure-preserving extractor; and Sarvam OCR is the lowest-latency option for structured workloads with known category caveats.

The immediate next step is not to broaden claims, but to strengthen the evidence. The highest-value follow-on work is expanding human-verified ground truth beyond forms so that future revisions can test the same operational hypotheses without consensus-tier circularity.

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TABLE VI

SUPPORTED PAIRWISE WILCOXON TESTS ON PER-DOCUMENT F1. CONSENSUS-INCLUSIVE ROWS EXCLUDE MISTRAL DUE TO CONSENSUS-GT CIRCULARITY. FORMS-ONLY ROWS USE THE HUMAN-VERIFIED $n = 20$ SUBSET; TESSERACT AND DOCLING EACH SIGNIFICANTLY OUTPERFORM MISTRAL ($p < 0.01$), WHILE TESSERACT VS. DOCLING IS NOT SIGNIFICANT ON FORMS ($p = 0.498$).

Comparison	n	p	**Winner**	**Effect size**
Consensus-inclusive (Mistral excluded)				
Tesseract vs. Surya	45	< 0.001	Surya	large ($d = 0.850$)
Tesseract vs. Docling	42	< 0.001	Docling	medium ($d = 0.576$)
Tesseract vs. Sarvam OCR	27	0.1482	Sarvam OCR	small ($d = 0.366$)
Surya vs. Docling	42	0.0241	Surya	small ($d = 0.314$)
Surya vs. Sarvam OCR	27	0.0906	Surya	small ($d = 0.448$)
Sarvam OCR vs. Docling	25	0.5424	Sarvam OCR	negligible ($d = 0.098$)
Forms-only (human-verified GT)				
Tesseract vs. Mistral OCR	16	0.0092	Tesseract	medium ($d = 0.729$)
Docling vs. Mistral OCR	16	0.0076	Docling	large ($d = 0.848$)
Tesseract vs. Docling	20	0.4980	Tesseract	negligible ($d = 0.059$)

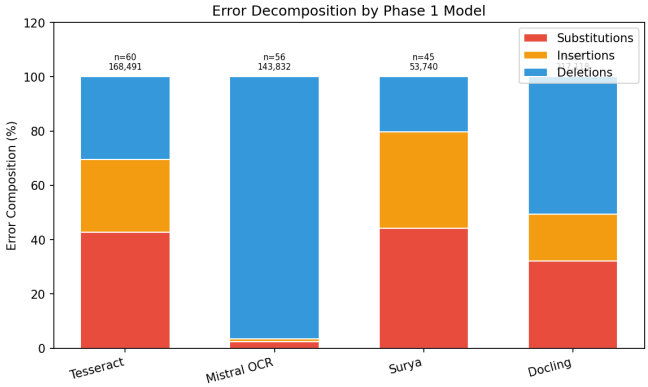

Fig. 4. Error decomposition across GT-evaluable documents. Bars show substitution, insertion, and deletion shares of total edit operations. Mistral is included descriptively, but not as a consensus-ground-truth ranking target because of circularity.

TABLE VII

TESSERACT LANGUAGE-PACK ABLATION ON 15 USGOV PDF FORMS. PAIRED DELTAS ARE TAKEN DOC-BY-DOC (MULTI-LANG MINUS ENG-ONLY). CHARACTER-LEVEL ACCURACY IS ESSENTIALLY UNCHANGED ON PURE-ENGLISH FORMS, BUT THE MULTI-LANGUAGE PACK MORE THAN DOUBLES LATENCY.

Variant	**CER**	**WER**	**F1**	**Latency (ms)**
`eng` only	0.3410	0.4386	0.9185	21,119
`eng+hin+tel+tam+ben`	0.3259	0.4297	0.9170	47,248
Paired Δ (multi $-$ eng)	-0.0151	-0.0089	-0.0015	$+26{,}129$